

RENE BENJAMIN ULLOA

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EDUCATION

Master of Arts | *Mathematics (Mathematics of Finance Specialization)*

September 2025 – Expected December 2026

Columbia University

New York, NY

• **Relevant Graduate Courses** – Stochastic Processes, Non-Linear Options Pricing, Statistical Inference, Numerical Methods

Master of Arts | *Economics*

August 2020 – December 2021

Duke University

Durham, NC

• **Relevant Graduate Courses** – Measure & Integration, Smooth Manifolds, Mathematical Finance, Continuous Time Methods, Applied

Econometrics in Microeconomics, Functional Analysis, Ordinary & Partial Differential Equations

• **Paper** – *Migration Movements Since 2000 & Its Influences*, with Xiaolin Ding

Bachelor of Science | *Mathematics & Economics*

August 2015 – May 2020

George Mason University

Fairfax, VA

• **Relevant Courses** – Real Analysis I-II, Ordinary Differential Equations (Ph.D.), Partial Differential Equations (Ph.D.), Topology of Metric

Spaces (Ph.D.), Advanced Linear Algebra, Macroeconomics, Microeconomics, Game Theory, Econometrics, Mathematical Statistics

PROFESSIONAL EXPERIENCE

Portfolio Implementation & Trading Summer Analyst

June 2026 – Present

AQR Capital Management

Greenwich, CT

- Designed and built an end-to-end quantitative framework to detect, decompose, and monitor latent thematic exposures (AI, tariffs, energy transition) in a systematic L/S equity fund, surfacing portfolio sensitivities that a multi-hundred-factor risk model classifies as residual noise.
- Developed a multi-level Transfer Coefficient decomposition of portfolio-versus-thematic-basket correlation, isolating pure thematic alignment from sector, market-beta, factor, and stock-specific overlap, plus a GLS-based attribution of which alpha signals drive each exposure and through which risk channel, enabling targeted hedging without degrading alpha.
- Shipped an interactive Streamlit dashboard (Python, Polars, Plotly, Parquet caching) delivering ex-ante exposure analytics plus ex-post realized-beta and P&L attribution with anomaly detection, replacing an ad-hoc manual process with continuous automated diagnostics for portfolio managers.

Fixed Income Data Scientist (Vice President)

August 2025 – April 2026

Tradeweb Markets LLC

New York, NY

- Architected a model-agnostic, agentic LLM chatbot (Docker, Linux) with custom hooks/skills, MCP integrations, and a retrieval-augmented knowledge base over Snowflake and Kafka, delivering responses and charts via Outlook SMTP to C-suite, clients, and developers.
- Engineered MCP (Model Context Protocol) integrations for automated report delivery with dynamic charts, and built a speech-to-text agent for the sales desk to accelerate dealer outreach on RFQ+ utilization and client rankings.
- Built and operated the fixed income data platform powering cover-policy models and dealer stream monitoring across MBS, MBSP, USDIRS, CDXIG, and Treasury markets, tracking spread tightness, ladder sizes, and platform comparisons; orchestrated with Prefect and GitLab CI/CD, with Tableau Server extraction via tabcmd and Python.
- Designed and published a cross-product liquidity index tracking 1-minute transaction cost standard deviations across UST, USDIRS, MBS FNCL, and CDXIG, with composite width analytics and macro-event overlays for market structure research.

Data & Insights Consultant (Data Science)

June 2024 – August 2025

Visa

Miami, FL

- Engineered Airflow-orchestrated ETL pipelines on Hadoop, Hive, and Spark, integrating cross-border, credit, and debit transaction sources to support analytics and machine learning initiatives for external clients and internal stakeholders.
- Productionized statistical metrics, machine learning anomaly detection models, and an automated email alerting system surfacing authorization-rate and transaction-pattern anomalies across domestic and international markets.

Research Analyst - TMT, Semiconductors, & Industrials

August 2022 - June 2024

Point72 Asset Management, L.P.

New York, NY

- Built Airflow-orchestrated ETL and modeling pipelines on credit card, satellite imagery, email receipt, job postings, geolocation, trade, and web scrape data, expanding the scope of actionable insights for PMs across Long/Short Equity and other investment strategies.
- Engineered predictive models using Stochastic Gradient Boosting, AdaBoost, and other ensemble methods, materially improving KPI forecast accuracy across TMT, Semiconductors, and Industrials coverage.
- Designed portfolio strategy analyses using backtesting frameworks, identifying critical investment performance drivers including factor returns, risk exposure, and turnover.

Equities Research Associate - Semiconductors, Hardtech, & Consumer Electronics

December 2021 - July 2022

M Science

New York, NY

- Built ETL pipelines in PySpark and SQL on Databricks transforming email receipt, credit card, trade, and web scrape data into modeling-ready datasets feeding KPI prediction APIs; developed statistical models and REST API endpoints serving research and client-facing systems.

SKILLS

Programming: Python (Pandas, NumPy, SciPy, Seaborn, Sklearn, TensorFlow), PySpark, TypeScript, SQL, R, MATLAB, C++

Data Engineering & Cloud: ETL/ELT pipelines (batch and streaming), data lake / lakehouse patterns, orchestration (Airflow, Prefect), CI/CD (GitLab, GitHub Actions), testing (pytest), schema design, REST APIs (Flask); Hive, Hadoop, Spark, Databricks, Delta Lake, Snowflake, Parquet, Kafka, AWS (S3, EC2, Glue), Docker, Kubernetes, Linux, Bloomberg

AI & LLM: Claude Code, Claude Agent SDK, MCP (Model Context Protocol), OpenAI Codex, RAG, agentic systems (custom hooks/skills/tools), model-agnostic chatbots, speech-to-text, prompt engineering

Statistical & ML: Time-series and panel data analysis, gradient boosting (e.g., XGBoost), random forests, logistic regression, anomaly detection, feature engineering, backtesting

Visualization: Tableau, Power BI, Streamlit, NiceGUI, Excel, $\LaTeX$

Languages: English (Native), Spanish (Native), Portuguese (Native), Italian (Intermediate), French (Basic)

Interests: Earnings Season, MMA, Judo, Freestyle Wrestling, Olympic Weightlifting, Neovim, Semiconductors